Calcurion Maths Worksheets

A-Level Further Mathematics

Topic: Further Mechanics 1 – Momentum & Impulse

Medium & Hard • 25 Questions

Scan for free resources

Section A – Medium (Q1–15)

1. A particle of mass 2 kg moving at $5\,\mathrm{m\,s^{-1}}$ receives an impulse of $6\,\mathrm{N\,s}$ in the direction of motion. Find its new speed. **[3]**

2. A particle of mass 3 kg has velocity $4\mathbf{i} - 2\mathbf{j}\,\mathrm{m\,s^{-1}}$. An impulse $2\mathbf{i} + 5\mathbf{j}\,\mathrm{N\,s}$ acts. Find the new velocity. **[3]**

3. Two particles of masses 2 kg and 3 kg collide directly. Speeds before impact are $4\,\mathrm{m\,s^{-1}}$ and $2\,\mathrm{m\,s^{-1}}$ towards each other. After impact they coalesce. Find the common speed. **[4]**

4. Define the coefficient of restitution e. **[2]**

5. A sphere hits a smooth vertical wall at $10\,\mathrm{m\,s^{-1}}$ at 30° to the normal. If $e = 0.5$, find the speed after impact. **[5]**

6. Particles of mass m and $2m$ approach each other at speeds $2u$ and u. Given $e = \frac{1}{2}$, find the speeds after impact. **[5]**

7. A ball is dropped from height h onto a horizontal floor with $e = \frac{1}{2}$. Find the height of the first rebound. **[4]**

8. An impulse $\mathbf{J}$ changes the velocity of a particle of mass m from $\mathbf{u}$ to $\mathbf{v}$. Write the impulse-momentum equation. **[2]**

9. A particle of mass 4 kg moving at $3\,\mathrm{m\,s^{-1}}$ collides with a stationary particle of mass 2 kg. After impact the first particle continues at $1\,\mathrm{m\,s^{-1}}$ in the same direction. Find e and the speed of the second particle. **[5]**

10. Find the loss in kinetic energy when two particles of masses m and $2m$ collide and coalesce, approaching at u and $2u$. **[4]**

11. A particle strikes a smooth plane at angle α to the normal with speed u. Coefficient of restitution e. Find the angle of rebound. **[4]**

12. Two equal masses collide directly with speeds u and 0. Show that if $e = 1$ the first stops and the second moves at u. **[3]**

13. A bullet of mass $0.02\,\text{kg}$ embeds in a block of mass $2\,\text{kg}$ at rest. Bullet speed $400\,\text{m}\,\text{s}^{-1}$. Find the common speed and KE lost. **[5]**

14. State Newton's experimental law of impact. **[2]**

15. A particle of mass m moving at u hits a stationary particle of mass km. Find the velocities after impact in terms of u, k and e. **[5]**

Section B – Hard (Q16–25)

16. A sphere hits a smooth wall at $20\,\text{m}\,\text{s}^{-1}$ at 45° to the wall. If $e = 0.6$, find the velocity after impact. **[5]**

17. Two particles collide obliquely. Masses $2\,\text{kg}$, $3\,\text{kg}$; velocities before $3\mathbf{i} + \mathbf{j}$ and $-\mathbf{i} + 2\mathbf{j}$. The line of centres is parallel to $\mathbf{i}$, $e = \frac{1}{2}$. Find velocities after impact. **[7]**

18. A ball is projected from ground level at speed u at angle α. It hits a smooth vertical wall at the top of its trajectory and rebounds with $e = \frac{1}{2}$. Find where it lands relative to the wall. **[6]**

19. Three particles A, B, C in a straight line. A hits B, then B hits C. Masses equal, $e = 1$ throughout, A has speed u, B and C at rest. Find final speeds. **[5]**

20. A particle of mass m moving at u collides with a stationary particle of mass m. Show that the maximum loss in KE occurs when $e = 0$. **[4]**

21. An impulse J acts on a particle of mass m at rest, at an angle θ to the horizontal. The particle is on a smooth horizontal table. Find the resulting velocity. **[3]**

22. Two spheres of equal mass and equal radius collide with line of centres at 30° to the velocity of the first (second at rest). Given $e = 1$, find the velocities after impact. **[6]**

23. A chain of collisions: n equal masses in a line, first has speed u, others at rest, $e = 1$. Describe the final motion. **[4]**

24. A particle hits a rough floor ($e = 0$, friction impulse also acts). Discuss the effect of friction on the horizontal component. **[4]**

25. Prove that for a direct impact of two spheres, the loss in KE is $\frac{1}{2}\frac{m_1 m_2}{m_1+m_2}(1-e^2)(u_1-u_2)^2$. **[5]**

End of Worksheet • *Show all working clearly. Take* $g = 9.8\,\mathrm{m\,s^{-2}}$ *unless otherwise stated.*